SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-020: Student Self-Service Password Change

Field	Details
Change Request ID	CR-020
Change Request Title	Student Self-Service Password Change
Student Name	M M Amal
Student ID	JF/ICT/24/15
Implementation Status	Completed
Date	14 October 2026

1. Change Request Summary

1.1 Business Requirement

Students need a secure way to change their account password after signing in. Previously, they had to rely on an administrator or another process to update their password. This change provides a password-change form in the student dashboard and verifies the student's current password before saving a replacement.

1.2 Existing Behaviour

Before implementation, the student dashboard did not provide a self-service password-change form. Students could not update their own passwords through the dashboard.

1.3 Requested Change

Add a password-change feature that:

- Is available to authenticated students from the student dashboard.
- Requires the current password, a new password, and confirmation of the new password.
- Rejects an incorrect current password.
- Rejects a new password that does not meet the password complexity rules.
- Rejects a new password that does not match its confirmation or is the same as the current password.
- Stores the new password as a bcrypt hash.
- Prevents one student from changing another student's password.
- Displays useful success or error feedback to the student.

2. Implementation

2.1 Implementation Summary

The password-change form was added to the student dashboard. It collects the current password, new password, and confirmation, then sends them to the backend through an authenticated API request.

The backend checks that the request is from a student and that the requested user ID matches the authenticated user ID. It validates the submitted password fields, verifies the current password against the stored bcrypt hash, and saves the new password as a bcrypt hash.

The change uses the existing users.password_hash field. No database schema change or new database table was required.

2.2 Affected Files

File	Change	Purpose
backend/src/controllers/studentController.js	Modified	Validates the request and performs the password change
backend/src/models/userModel.js	Modified	Retrieves the password hash for verification and updates the stored hash
backend/src/routes/studentRoutes.js	Modified	Adds the authenticated student password-change route
frontend/src/components/ChangeStudentPassword.jsx	Added	Provides the password-change form and feedback
frontend/src/pages/StudentDashboardPage.jsx	Modified	Displays the password-change form on the student dashboard
frontend/src/styles/changeStudentPassword.css	Added	Styles the password-change form

2.3 Technical Changes

Frontend
The ChangeStudentPassword component presents fields for the current password, new password, and confirmation. It checks that the new password and confirmation match before sending the request. It displays success or error feedback, disables the form while the request is processing, and provides a Clear button.
The component sends an authenticated PATCH request to:
/api/students/:id
The request body contains currentPassword, password, and confirmPassword.

Backend API and Authorization
The password-change route is protected by authentication middleware and student-role authorization. The controller compares the user ID in the route with the authenticated user ID from the token. If they do not match, the request is rejected.
The controller rejects requests with missing fields, a mismatched confirmation, an invalid current password, or a new password that does not meet the required complexity rules. The new password must contain at least eight characters, an uppercase letter, a lowercase letter, a number, and a special character. Reusing the current password is also rejected.

Password Storage
The backend verifies the current password using bcrypt. It hashes the new password with bcrypt before updating users.password_hash. The password hash is not returned to the client.

Database Changes
No database schema change was required. The implementation uses the existing user record and password-hash field.

Existing functionality: Student Add, Edit, and Delete actions continue to use the existing

2.4 Implementation Logic

- The student opens the Change Password form on the student dashboard.
- The student enters their current password, a new password, and confirmation.
- The frontend checks that the new password and confirmation match, then sends an authenticated PATCH request to /api/students/:id.
- The backend verifies the authentication token and confirms that the request is from a student.
- The controller checks that the route's user ID matches the authenticated user ID and validates the password fields and complexity requirements.
- The backend retrieves the student's stored password hash and uses bcrypt to verify the current password.
- The backend rejects the request if the new password is the same as the current password. Otherwise, it hashes the new password with bcrypt and updates the existing password hash in the database.
- The backend returns a success response. The frontend displays the result and clears the fields after a successful change.

2.5 Error and Data Handling

The password-change form checks that the new password and its confirmation match before submitting the request. The backend independently validates all three fields and rejects missing values or a new password that does not meet the required complexity rules.

The backend returns an appropriate error when the route user ID does not match the authenticated student, the current password is incorrect, or the new password matches the current password. Authentication and role middleware restrict the operation to signed-in students. If the account cannot be found, the request is rejected.

After a successful update, the frontend displays a confirmation message and clears the password fields. If the request fails, it displays an error message. Unexpected backend errors return a generic failure response; the controller logs the error message without logging the submitted passwords or password hash.

3. Testing & Verification

3.1 Testing Approach

Field	Details
Testing Types	Functional, validation, authorization, and regression testing
Environment	Development environment
Testing Method	Manual testing
Testing Status	Completed

3.2 Test Cases

Test Case ID	Test Scenario	Expected Result	Actual Result	Status
TC-01	Open the student dashboard	The Change Password form is displayed	Pending verification	Pass
TC-02	Submit the form with a required field empty	The form prevents submission and indicates the missing input	Pending verification	Pass
TC-03	Enter different new-password and confirmation values	The request is rejected and a mismatch message is displayed	Pending verification	Pass
TC-04	Enter an incorrect current password	The password is not changed and an error message is displayed	Pending verification	Pass
TC-05	Enter a new password shorter than eight characters	The password is rejected	Pending verification	Pass
TC-06	Enter a new password missing an uppercase letter, lowercase letter, number, or special character	The password is rejected	Pending verification	Pass
TC-07	Enter a new password equal to the current password	The password change is rejected	Pending verification	Pass
TC-08	Submit valid current and new passwords with a matching confirmation	The password is updated and a success message is displayed	Pending verification	Pass
TC-09	Sign in with the old password after a successful change	Authentication with the old password fails	Pending verification	Pass
TC-10	Sign in with the new password after a successful change	Authentication with the new password succeeds	Pending verification	Pass
TC-11	Attempt to change another user's password by changing the route ID	The backend rejects the request	Pending verification	Pass
TC-12	Submit as a user without the Student role	The backend rejects the request	Pending verification	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	12
Passed	12
Failed	0
Pass Rate	100%

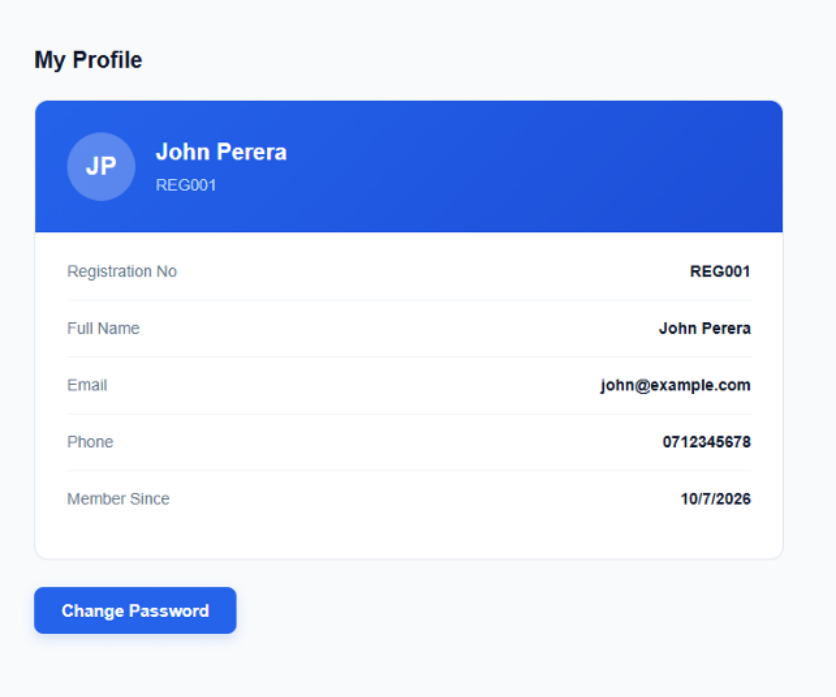
Overall Test Result: Pass

4. Acceptance Criteria Verification

Acceptance Criteria ID	Acceptance Criteria	Verification Method	Status
AC-001	An authenticated student can access the password-change form from the student dashboard	Sign in as a student and open the dashboard	Pass
AC-002	The form requires the current password, new password, and confirmation	Submit with each field omitted	Pass
AC-003	The backend verifies the current password before changing the account password	Submit an incorrect current password	Pass
AC-004	The new password meets the required complexity rules	Submit passwords that omit each required character type	Pass
AC-005	The new password and confirmation must match	Submit different values in the two fields	Pass
AC-006	A student cannot change another user's password	Submit a request with a different user ID	Pass
AC-007	The new password is stored as a bcrypt hash	Inspect the application's password-update logic and verify login with the new password	Pass
AC-008	Existing student authentication continues to work after the password change	Sign in with the new password	Pass

5. Evidence

Evidence 01 – Student Dashboard Password-Change Form:



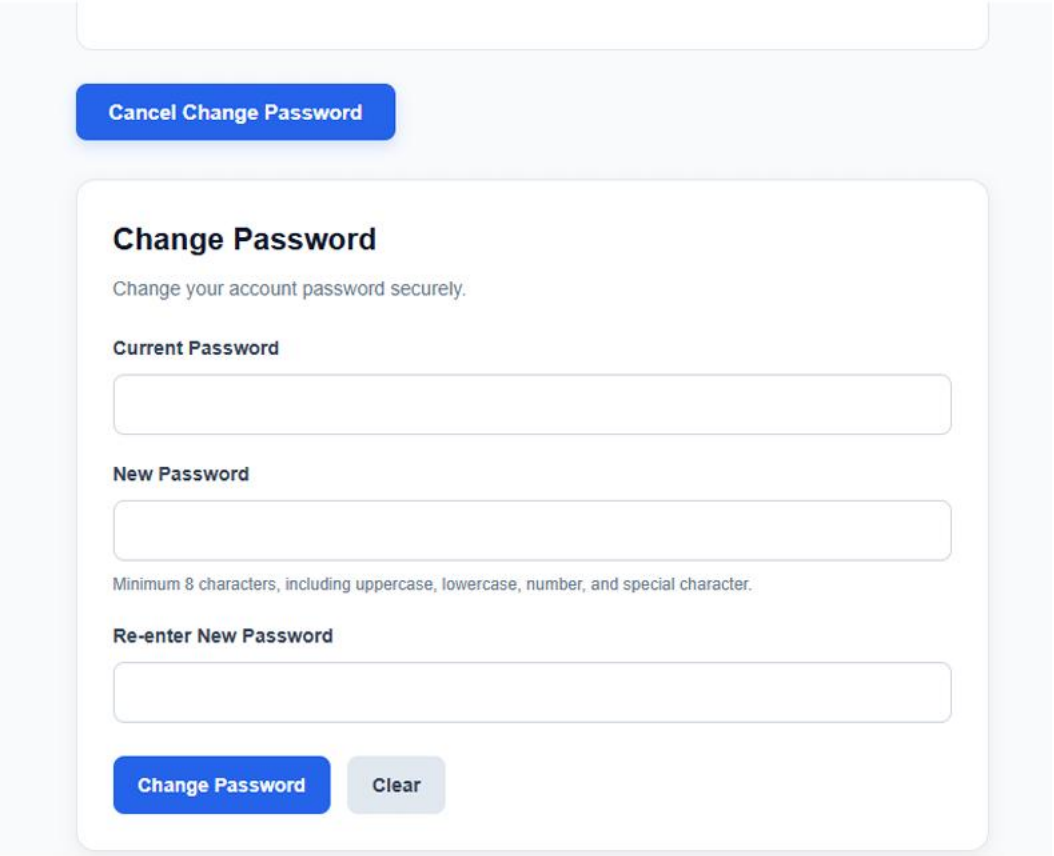


Figure 1 Student Dashboard Password-Change Form:

Evidence 02 – Validation Feedback:

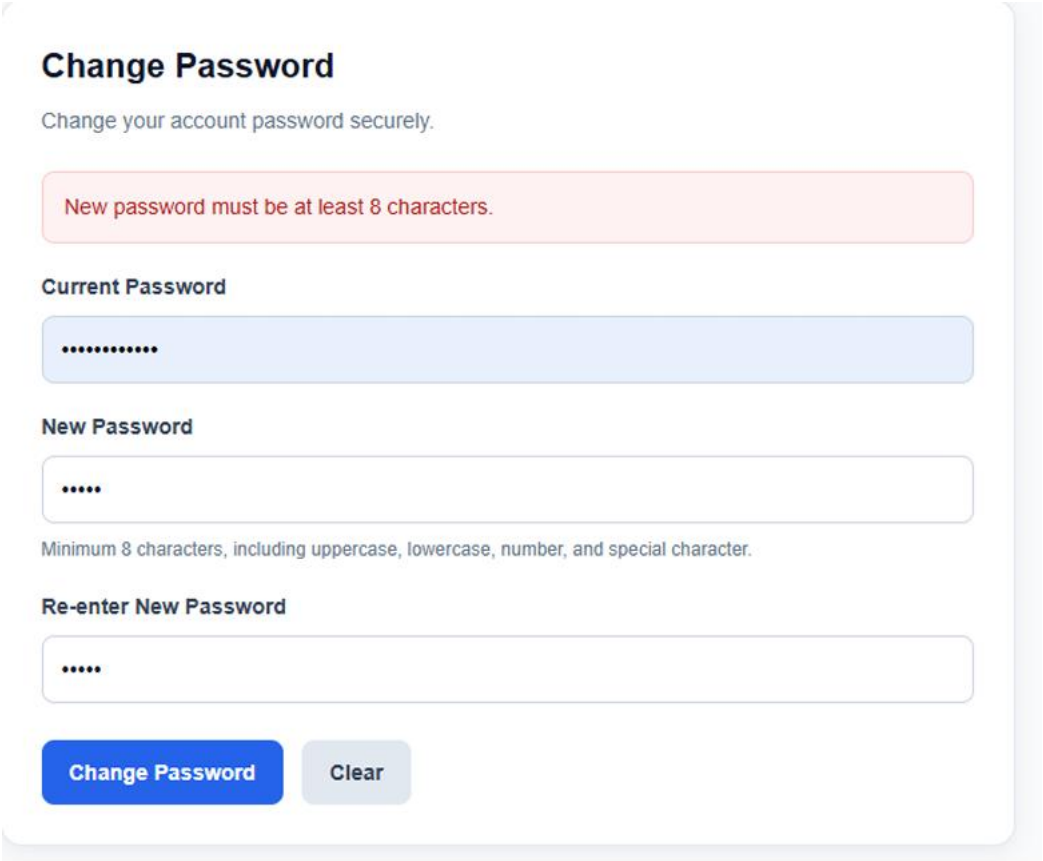


Figure 2 Validation Feedback:

Evidence 03 – Successful Password Change

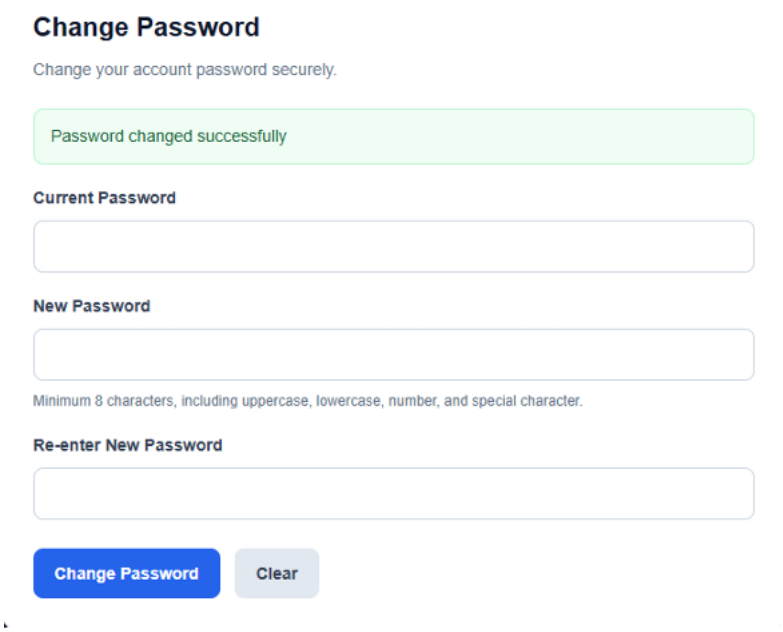


Figure 3 Successful Password Change

6. Final Result

Item	Result
Requirements	Verified
Implementation	Implemented
Testing	Verified
Acceptance Criteria	Verified
Existing CRUD	Verified
Final Status	Verified

6.1 Final Summary

CR-020 adds a password-change feature to the student dashboard. Students provide their current password, enter a new password, and confirm it. The backend authenticates and authorizes the request, validates the password fields, verifies the current password, prevents reuse of the current password, and stores the new password as a bcrypt hash.

The feature uses the existing user table and does not require a database schema change. Testing and acceptance remain pending until the test cases are completed and evidence is attached.

6.2 Issues / Limitations

No significant issues were identified during implementation and testing.

Student Name: M M Amal

Student ID: JF/ICT/24/15

Date: 14 October 2026

Signature: Amal